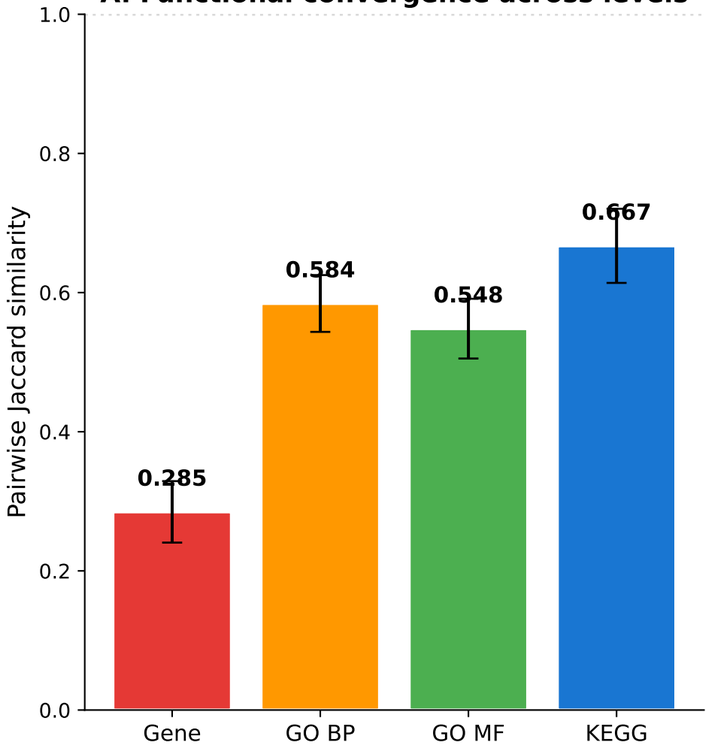
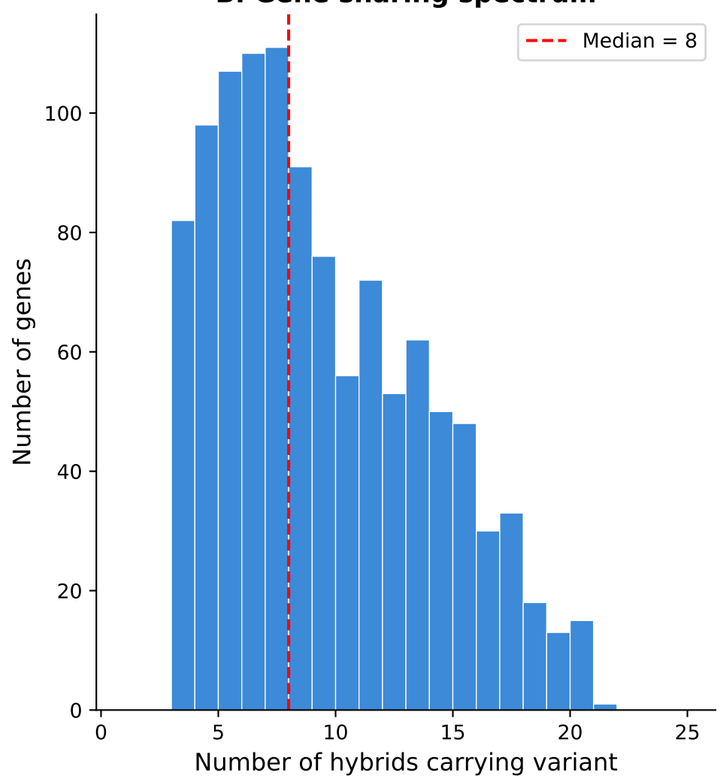
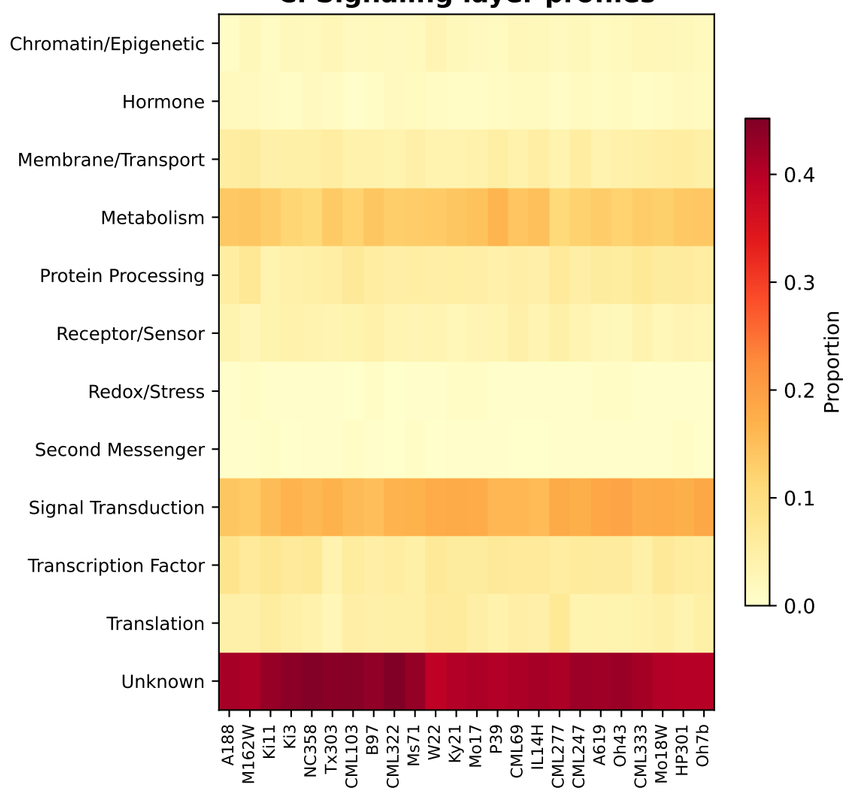
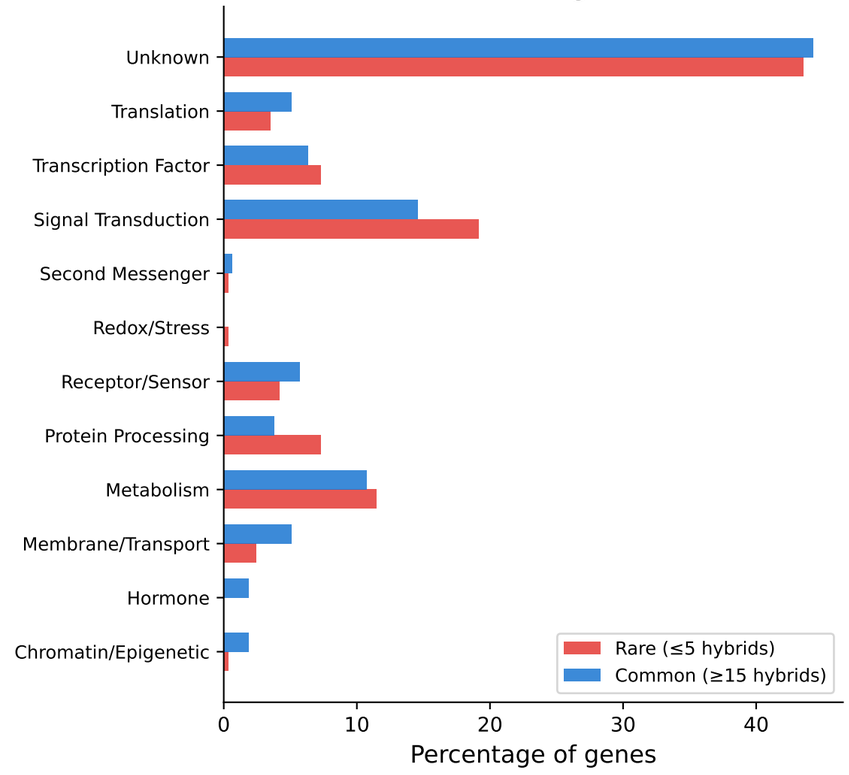
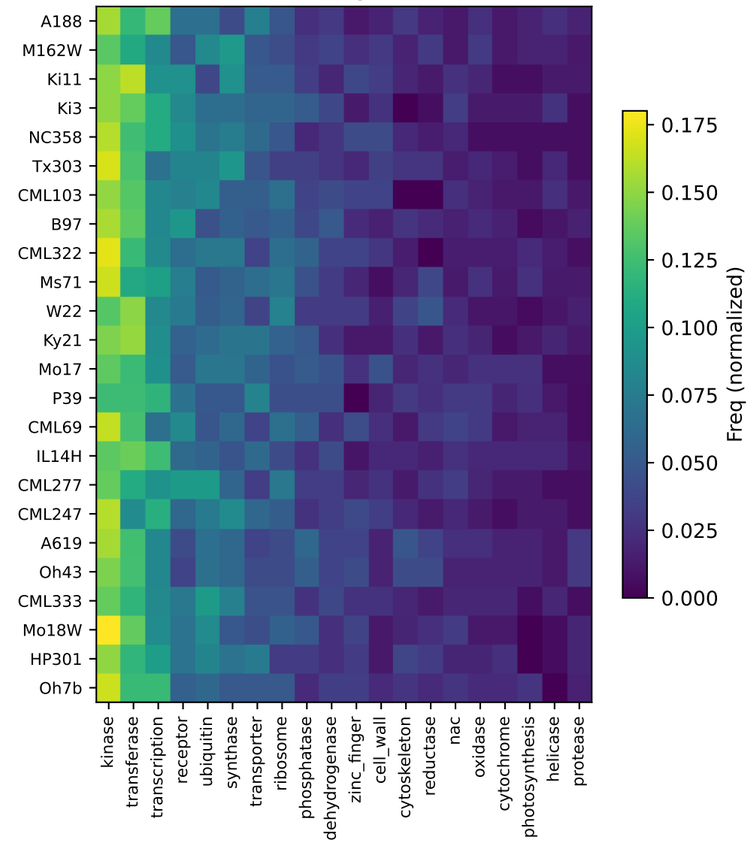
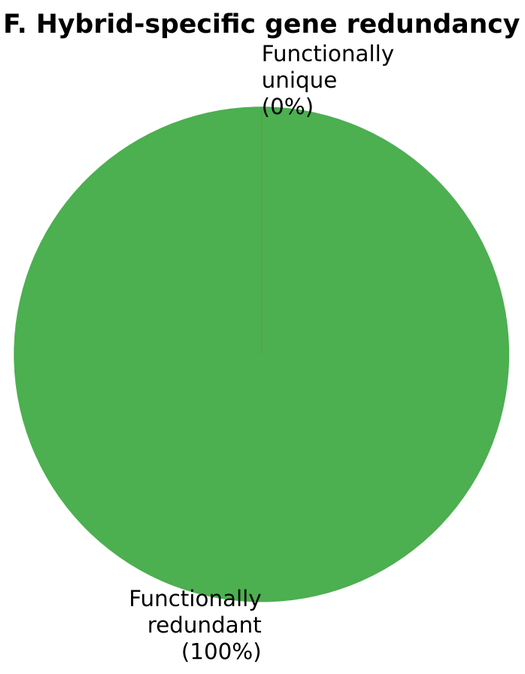

A. Functional convergence across levels
Pairwise Jaccard similarity
1.0
0.8
0.6
0.4
0.2
0.0
0.285
0.584
0.548
0.667
Gene
GO BP
GO MF
KEGG


B. Gene sharing spectrum
Median = 8
Number of genes
100
80
60
40
20
0
0
5
10
15
20
25
Number of hybrids carrying variant


C. Signaling layer profiles
Chromatin/Epigenetic
Hormone
Membrane/Transport
Metabolism
Protein Processing
Receptor/Sensor
Redox/Stress
Second Messenger
Signal Transduction
Transcription Factor
Translation
Unknown
A188
M162W
Ki11
Ki3
NC358
Tx303
CML103
B97
CML322
Ms71
W22
Ky21
Mo17
P39
CML69
IL14H
CML277
CML247
A619
Oh43
CML333
Mo18W
HP301
Oh7b
Proportion
0.4
0.3
0.2
0.1
0.0


D. Rare vs common gene functions
Unknown
Translation
Transcription Factor
Signal Transduction
Second Messenger
Redox/Stress
Receptor/Sensor
Protein Processing
Metabolism
Membrane/Transport
Hormone
Chromatin/Epigenetic
Rare (≤5 hybrids)
Common (≥15 hybrids)
0
10
20
30
40
Percentage of genes


E. Functional keyword profiles
A188
M162W
Ki11
Ki3
NC358
Tx303
CML103
B97
CML322
Ms71
W22
Ky21
Mo17
P39
CML69
IL14H
CML277
CML247
A619
Oh43
CML333
Mo18W
HP301
Oh7b
kinase
transferase
transcription
receptor
ubiquitin
synthase
transporter
ribosome
phosphatase
dehydrogenase
zinc_finger
cell_wall
cytoskeleton
reductase
nac
oxidase
cytochrome
photosynthesis
helicase
protease
Freq (normalized)
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000


F. Hybrid-specific gene redundancy
Functionally unique (0%)
Functionally redundant (100%)